

Neighborhood Quality and Screen Use: Findings From the Adolescent Brain Cognitive Development Study



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ABSTRACT

OBJECTIVE: To determine the relationship between various measures of neighborhood quality and adolescent screen use in a demographically diverse, US cohort of early adolescents.

METHODS: We analyzed cohort data of 8446 adolescents (aged 9–10 at baseline in 2016–2018) with 2 years of follow-up from the Adolescent Brain Cognitive Development (ABCD) Study. Mixed-effects models adjusted for age, sex, race and ethnicity, household income, parents' highest education, and study site were used to examine the associations of baseline indices of children's neighborhood quality measures based on geospatial location data with daily screen time measures at baseline, Year 1, and Year 2.

RESULTS: Lower child opportunity index (COI), higher social vulnerability index (SVI), and higher area deprivation index (ADI) at baseline were associated with higher total screen time, particularly higher television/movie watching, video game playing, and video streaming. Specifically, compared to the

highest COI quintile, lower COI quintiles (1st–4th) were associated with higher total screen time ($B = 0.39$ – 0.64 , $P < .05$). Compared to the first SVI quintile, higher SVI quintiles (2nd–5th) were associated with higher total screen time ($B = 0.28$ – 0.64 , $P < .01$). Similarly, compared to the first ADI quintile, higher ADI quintiles (2nd–5th) were also associated with higher total screen time ($B = 0.46$ – 1.17 , $P < .001$).

CONCLUSIONS: Overall, this study found that neighborhood quality is associated with higher screen time, over and above other factors, suggesting it is important to consider the role of environmental factors and identify potential targets for intervention at the neighborhood level.

KEYWORDS: built environment; environmental health; neighborhood; screen time; social media

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WHAT'S NEW

In a demographically diverse cohort of 8446 early adolescents, lower child opportunity index, higher social vulnerability index, and higher area deprivation index were associated with higher total screen time, particularly higher television/movie watching, video game playing, and video streaming.

Aspects of neighborhood quality, including educational opportunities, access to healthy food and green spaces, and socioeconomic conditions, play a crucial role in shaping child development and health behaviors, including screen use.^{1,2} These neighborhood qualities have been shown to be associated directly and indirectly (via

parenting stress and adverse childhood experiences) with adolescents' behavior problems and social skills, and indirectly with their delinquency.³ Previous studies have suggested that neighborhood qualities are a key determinant of health and well-being and should be considered alongside sociodemographic factors to better understand child development.^{1,2}

Studies have demonstrated that household and individual-level factors, such as Black/multiracial race and lower parental education are associated with higher adolescent screen time, or time spent on various screen modalities, including viewing television (TV) shows or movies, watching videos (eg, YouTube), playing video games, texting, video chatting (eg, Skype, FaceTime), and using social media (eg, Facebook, Instagram, Twitter/X).⁴

Over and above that, other studies have shown that at the neighborhood level, access to outdoor activities, low social disorder, and favorably perceived aesthetics (defined as “nice things to look at”) are associated with decreased screen use.^{5–7} Excessive screen time in adolescents is linked to adverse outcomes, such as increased sedentary behavior, heightened risk of depression and anxiety, disordered eating, and reduced quality of life.^{8–10}

Although previous literature has reported on the relationship between certain neighborhood factors and both child development and total screen use, there is a paucity of studies investigating the associations between neighborhood factors and specific types of media use, such as social media, particularly in large, demographically diverse samples. Understanding the relationships between neighborhood factors and problematic screen use patterns in children has become increasingly important, given the rise of social media use—a distinct category of screen time—in early adolescents, the need for early intervention in high-risk children, and the role of policy-driven environmental safety measures, such as programs that create nurturing, family-friendly environments, build community resilience, and reinforce social support in vulnerable neighborhoods.^{3,11}

This study aims to quantify the associations between neighborhood quality indices (metrics of neighborhood opportunity, vulnerability, and disadvantage derived from geospatial location data) and adolescent screen time patterns using comprehensive data from a diverse national sample in the United States. Additionally, we seek to identify key environmental predictors of excessive screen use to inform public health interventions. Based on prior research, we hypothesize that lower neighborhood quality will be associated with higher screen use among adolescents.

METHODS

STUDY POPULATION

We utilized data from baseline through Year 2 follow-up of the Adolescent Brain Cognitive Development (ABCD) Study (5.1 release). The ABCD Study is an ongoing prospective cohort study of health and cognitive development with 11,875 participants aged 9 to 10 years at baseline (2016–2018), recruited from 21 recruitment sites across the United States. The participants, recruitment, protocol, and measures of the ABCD Study are comprehensively described elsewhere.¹² Among 11,875 participants, 2368 had missing screen time data, 724 had missing sociodemographic data, and 424 had missing data for at least one index of children’s neighborhood opportunity or quality, leaving 8446 participants for the current analysis (Appendix A). Centralized institutional review board approval was granted by the University of California, San Diego and relied on by study sites. Written assent was obtained from participants, and written informed consent was provided by their caregivers.

MEASURES

NEIGHBORHOOD QUALITY

The ABCD Study uses geospatial location data incorporated by the Linked External Data Environmental workgroup to gather individual, neighborhood, and state-level information based on a child’s place of residence.¹³ The goal is to better understand the environmental contexts in which today’s children grow up.¹³ Using linked external data, we analyzed the following 3 indices of neighborhood quality.

Child Opportunity Index (COI). The COI 2.0 is a national metric of neighborhood opportunity, combining 29 indicators of neighborhood conditions for 72,000 census tracts in the United States. The ABCD Study includes scores for the overall COI and the following 3 domains that comprise the overall COI: (1) education (eg, third-grade reading and math proficiency, school poverty), (2) health and environment (eg, access to healthy food and green spaces), and (3) social and economic opportunities.¹⁴ To create nationally normed COI levels for the overall COI index and each of the 3 COI domains, all neighborhoods were ranked nationwide and divided into quintiles.¹⁵ The first quintile represents the lowest neighborhood opportunity, and the fifth quintile represents the highest neighborhood opportunity.

Social Vulnerability Index (SVI). The SVI, developed by the Centers for Disease Control and Prevention (CDC), is a composite measure designed to identify communities most at risk from stressors like natural disasters, disease outbreaks, and human-caused events.¹⁶ The SVI includes 15 variables from the American Community Survey (ACS), organized into 4 themes as follows: socioeconomic status (1–4), household composition and disability (5–8), minority status and language (9–10), and housing type and transportation (11–15). To calculate the SVI, percentiles were derived for each variable at the county or census-tract level, summed within each theme, and then further summed across all themes. Our study used the SVI at the census-tract level and categorized participants into quintiles based on the SVI national percentiles, with the first quintile indicating lower vulnerability to disasters and disease outbreaks and the fifth quintile indicating greater vulnerability.

Area Deprivation Index (ADI). The ADI is a composite measure of neighborhood disadvantage, representing socioeconomic status, developed by Singh.¹⁷ The ADI is derived from 17 socioeconomic indicators, including median home value and the percentage of families living below the poverty line. Further details on its calculation are available elsewhere.¹³ The composite ADI national percentiles were calculated at the census-tract level for participants’ addresses at baseline. The ADI data were sourced from the 2011 to 2015 American Community Survey (ACS). Participants were categorized into quintiles based on ADI percentiles, with the first quintile indicating the least disadvantaged

neighborhoods and the fifth quintile representing the most disadvantaged.

SCREEN TIME

Screen time was assessed using the ABCD Youth Screen Time Survey,¹⁸ where data were collected annually and harmonized across years. Adolescents answered 14 questions about the number of hours per weekday and weekend day spent on various screen modalities as follows: viewing TV shows or movies, watching videos (eg, YouTube), playing video games, texting, video chatting (eg, Skype, FaceTime), and using social media (eg, Facebook, Instagram, Twitter). We divided screen time into the following 4 categories: TV/movies, video games (single and multi-player), videos, and digital socializing (texting, video chatting, and social networking). We calculated the time spent on each screen modality for weekdays and weekends separately. Then, we computed a weighted average to determine participants' average daily time spent on each modality using the following formula: $[(\text{weekday average} \times 5) + (\text{weekend average} \times 2)]/7$.¹⁹ Total daily screen time was the weighted average of daily weekday and weekend screen time reported by participants.

COVARIATES

Potential sociodemographic confounders at baseline for the associations between indices of children's neighborhood quality and screen time were included. Age was measured in years, and sex was categorized as either female or male. Race and ethnicity were classified as Asian, Black, Latino/Hispanic, Native American, White, or other. Household income was grouped into the following 6 categories: less than \$25,000; \$25,000 to \$49,999; \$50,000 to \$74,999; \$75,000 to \$99,999; \$100,000 to \$199,999; and \$200,000 or more. Parental education level was classified as high school or less versus college or higher. Study site was also included as a confounding factor, accounting for potential geographic variation.

STATISTICAL ANALYSIS

We used baseline indices of children's neighborhood quality (COI, SVI, ADI) as primary predictors.¹³ Repeated measures of total screen time and screen time separated across 4 modalities from baseline, Year 1, and Year 2 were used as outcomes. Linear mixed-effects models with robust standard errors were used to assess the association of baseline indices of children's neighborhood quality (COI, SVI, ADI) with daily screen time measures at baseline and 2 follow-up years. Each year of data was treated as a separate observation nested within individuals, with a random intercept for each participant to account for within-subject correlation. Follow-up year was included as a fixed effect to model temporal trends. Thus, each participant contributed baseline and 2 follow-up years of screen time data, and the resulting estimates

represent average effects across follow-up while accounting for both within- and between-person variability. Associations between each COI domain and daily screen time measures were also assessed. All models were adjusted for the following baseline covariates: age, sex, race and ethnicity, household income, parent's highest education, and study site. ABCD propensity weights were applied to yield representative estimates based on the ACS from the US Census. A P -value $< .05$ was used to indicate statistical significance. Analyses were conducted using Stata 18 (StataCorp, College Station, Tex).

RESULTS

Characteristics of 8446 participants included in our study are shown in Table 1. Out of our sample, 48.4% were female, 41.9% were non-White, and 53.3% had a household income of less than \$75,000. Screen time characteristics are presented in Table 2. Screen time increased from 3.8 h/day at baseline to 6.7 h/day at Year 2, with the greatest screen time spent watching TV/movies at baseline and playing video games at Year 2.

Table 3 presents the adjusted models for the associations between baseline indices of children's neighborhood quality (COI, SVI, ADI) and total screen time, along with its subtypes, at baseline, Year 1, and Year 2. Lower COI, higher SVI, and higher ADI at baseline were associated with higher total screen time at baseline, Year 1, and Year 2. Specifically, compared to the highest COI quintile, lower COI quintiles (1st–4th) were associated with higher total screen time ($B = 0.39$ – 0.64 , $P < .05$). Compared to the first SVI quintile, higher SVI quintiles (2nd–5th) were associated with higher total screen time ($B = 0.28$ – 0.64 , $P < .01$). Similarly, compared to the first ADI quintile, higher ADI quintiles (2nd–5th) were also associated with higher total screen time ($B = 0.46$ – 1.17 , $P < .001$). Lower COI, higher SVI, and higher ADI were also generally associated with higher total time spent playing video games, watching TV/movies, and streaming videos. Higher ADI quintiles were associated with higher total time spent digitally socializing compared to the first ADI quintile.

Models for the associations between the 3 COI domains at baseline and total screen time and its subtypes at baseline, Year 1, and Year 2 are shown in Appendix B. Lower quintiles for the 3 COI domains were generally associated with higher total screen time.

DISCUSSION

In this demographically diverse, national sample of 8446 US adolescents from the ABCD study, we found that lower COI (total and across all 3 domains), higher SVI, and higher ADI were associated with higher total screen time, particularly higher TV/movie watching, video game playing, and video streaming, without a clear

Table 1. Sociodemographic Characteristics of ABCD Study Participants Included in This Analysis at Baseline (N = 8446)

Sociodemographic Characteristics	Mean (SD) / %
Age (y)	9.9 (0.6)
Sex	
Female	48.4%
Male	51.6%
Race and ethnicity	
Asian	5.4%
Black	14.0%
Latino/Hispanic	18.0%
Native American	3.2%
Other	1.3%
White	58.1%
Household income	
\$24,999 or less	15.3%
\$25,000 to \$49,999	19.8%
\$50,000 to \$74,999	18.2%
\$75,000 to \$99,999	14.3%
\$100,000 to \$199,999	24.7%
\$200,000 and greater	7.7%
Parent's highest education	
High school education or less	15.2%
College education or more	84.8%

ABCD indicates Adolescent Brain Cognitive Development. ABCD sampling weights were applied to yield representative estimates based on the American Community Survey from the US Census

Table 2. Average Daily Screen Time of ABCD Study Participants Included in This Analysis From Baseline to Year 2 (N = 8446)

Screen Time (h)	Mean (SD)		
	Baseline	Year 1	Year 2
Total screen time	3.8 (3.0)	4.6 (3.5)	6.7 (5.6)
Video games	1.1 (1.2)	1.3 (1.3)	2.1 (2.8)
TV shows or movies	1.3 (1.2)	1.3 (1.2)	1.6 (1.7)
Videos	1.0 (1.2)	1.2 (1.2)	1.4 (1.9)
Digital socializing	0.5 (1.2)	0.9 (1.7)	1.8 (3.1)

ABCD indicates Adolescent Brain Cognitive Development. ABCD sampling weights were applied to yield representative estimates based on the American Community Survey from the US Census

dose-response relationship. Findings with digital socializing were less strong and mixed. We hypothesize that these mixed findings reflect the greater dependence of digital socializing on individual and interpersonal factors, rather than structural or neighborhood-level variables. The association of lower neighborhood quality and higher total screen time for these mixed screen modalities, even when controlling for individual sociodemographic factors, suggests that the neighborhood context may contribute to increased screen use. Previous studies have found that poorer perceived aesthetics, higher neighborhood social disorder, and inadequate access to outdoor activities were associated with higher screen time in adolescents.^{5–7} Further, other studies that have looked at socioeconomic

status, but not neighborhood quality, have found that lower socioeconomic status, which is often correlated with lower neighborhood opportunity and higher vulnerability, is linked to higher screen time.²⁰

Our findings have significant implications for medicine, policy, and public health by informing an understanding of environmental influences on youth digital behaviors. With the rise of social media use—a distinct category of screen time—in early adolescents, these findings can help promote public health reform and digital wellness, especially through the targeting of areas with low COI, high SVI, and high ADI with public health campaigns. More screening and media literacy education can also be conducted for individuals in disadvantaged neighborhoods. Clinics in neighborhoods with low COI, high SVI, and high ADI may consider providing specific services to address screen time among adolescents as well.³ However, while targeted intervention may help in the short term, policymakers should consider how neighborhoods can be improved to prevent these downstream effects on adolescent screen time, such as increasing recreational spaces, access to job training, and extracurricular programming for adolescents.

Key strengths of this study include a large sample size, repeated measures of screen time, and a detailed assessment of various screen time modalities. Another strength is the use of comprehensive neighborhood data from ABCD using linked external data that capture children’s built and natural environments.¹³ However, the findings may have limited generalizability, as residential geospatial data may not fully capture where children spend their time or how much time they spend in different locations, and experiences can vary even within the same census tract,¹³ making it difficult to suggest specific, feasible recommendations. Further, mediators may include safety/lack of access to green space, mental health, stress, and adverse childhood experiences. Although potential confounders were adjusted for, there may be residual unmeasured confounding. Additionally, screen time measures were self-reported, introducing potential social desirability and recall biases.^{18,21,22} Lastly, while our study was longitudinal, we were unable to take into account adolescents who moved over the follow-up period due to data privacy restrictions, and the relatively short 2-year follow-up limits our ability to observe longer-term changes in screen time as influenced by neighborhood quality.

In conclusion, we found that lower neighborhood quality was associated with higher total screen time, particularly higher TV/movie watching, video game playing, and video streaming. By exploring longitudinal associations, further characterizing potential mediators of this relationship, and aiming to replicate these results using objective measures of adolescent screen time, future studies can build on our work to illuminate the effects of neighborhood quality on screen time.

Table 3. Associations of Baseline Indices of Children's Neighborhood Quality and Screen Time at Baseline, Year 1, and Year 2 in the ABCD Study (N = 8446)

Index of Children's Neighborhood Quality	Total Screen Time (h/day)		Video Games (h/day)		TV Shows or Movies (h/day)		Videos (h/day)		Digital Socializing (h/day)	
	Adjusted B (95% CI)	P	Adjusted B (95% CI)	P	Adjusted B (95% CI)	P	Adjusted B (95% CI)	P	Adjusted B (95% CI)	P
Child Opportunity Index quintile										
1st (lowest)	0.39 (0.05, 0.73)	.024	0.09 (−0.05, 0.24)	.209	0.10 (0.00, 0.21)	.061	0.10 (−0.02, 0.22)	.119	0.15 (−0.02, 0.32)	.085
2nd	0.64 (0.35, 0.94)	< .001	0.16 (0.03, 0.28)	.013	0.13 (0.04, 0.22)	.006	0.22 (0.12, 0.33)	< .001	0.17 (0.03, 0.31)	.020
3rd	0.59 (0.34, 0.84)	< .001	0.19 (0.08, 0.29)	.001	0.16 (0.08, 0.24)	< .001	0.23 (0.14, 0.32)	< .001	0.09 (−0.03, 0.21)	.125
4th	0.39 (0.20, 0.57)	< .001	0.10 (0.02, 0.18)	.012	0.11 (0.05, 0.16)	< .001	0.13 (0.06, 0.19)	< .001	0.07 (−0.01, 0.15)	.107
5th (highest)	reference		reference		reference		reference		reference	
Social Vulnerability Index quintile										
1st (lowest)	reference		reference		reference		reference		reference	
2nd	0.28 (0.09, 0.48)	.004	0.06 (−0.02, 0.14)	.152	0.08 (0.02, 0.13)	.011	0.09 (0.02, 0.16)	.013	0.12 (0.03, 0.21)	.010
3rd	0.48 (0.25, 0.71)	< .001	0.12 (0.03, 0.22)	.013	0.17 (0.10, 0.25)	< .001	0.14 (0.06, 0.22)	.001	0.10 (0.00, 0.20)	.060
4th	0.64 (0.38, 0.91)	< .001	0.16 (0.05, 0.28)	.005	0.12 (0.03, 0.20)	.005	0.24 (0.14, 0.34)	< .001	0.17 (0.04, 0.30)	.012
5th (highest)	0.43 (0.11, 0.75)	.009	0.09 (−0.05, 0.22)	.217	0.08 (−0.02, 0.18)	.120	0.15 (0.03, 0.27)	.013	0.16 (0.00, 0.32)	.060
Area Deprivation Index quintile										
1st (lowest)	reference		reference		reference		reference		reference	
2nd	0.46 (0.26, 0.65)	< .001	0.16 (0.08, 0.24)	< .001	0.13 (0.07, 0.18)	< .001	0.13 (0.07, 0.20)	< .001	0.07 (−0.01, 0.15)	.102
3rd	0.84 (0.59, 1.09)	< .001	0.27 (0.17, 0.38)	< .001	0.20 (0.13, 0.28)	< .001	0.29 (0.20, 0.37)	< .001	0.13 (0.02, 0.24)	.026
4th	1.17 (0.83, 1.51)	< .001	0.34 (0.19, 0.48)	< .001	0.25 (0.15, 0.36)	< .001	0.32 (0.21, 0.44)	< .001	0.36 (0.19, 0.53)	< .001
5th (highest)	1.05 (0.65, 1.44)	< .001	0.28 (0.11, 0.44)	.001	0.30 (0.18, 0.43)	< .001	0.28 (0.14, 0.43)	< .001	0.27 (0.07, 0.47)	.007

ABCD indicates Adolescent Brain Cognitive Development. Bold indicates $P < .05$. Models represent the abbreviated output from 15 mixed-effects linear regression models with baseline indices of children's neighborhood quality as the independent variables and repeated measures of total screen time and screen time modalities (baseline, Year 1, and Year 2) as the dependent variables. Models were adjusted for age, sex, race and ethnicity, household income, parent education, study site, and follow-up year. ABCD sampling weights were applied to yield representative estimates based on the American Community Survey from the US Census

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DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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AVAILABILITY OF DATA AND MATERIALS

Data used in the preparation of this article were obtained from the ABCD Study (<https://www.abcdstudy.org>), held in the NIMH Data Archive (NDA).

ADDITIONAL INFORMATION

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SUPPORTING INFORMATION

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